

assignment #01

Course: CHE 433: PROCESS SIMULATION WITH ASPEN PLUS
Term: Fall 2025
Professor: Dr. Michael Caracotsios
Office: EIB 244
Cell Phone: (630) 696-0934
E-mail: mcaracot@uic.edu

Student names: _____

Due Date: Monday, September 8th in Class (3:30PM)

INSTRUCTIONS FOR ASSIGNMENT DELIVERY

- Prepare a **typed** report of your assignment.
 - Include this cover page with your name(s) in the report.
 - Upload your report as a PDF file by the Due Date and Time on the UIC Blackboard.
 - Bring a hard copy of your report as a PDF file by the Due Date and Time in class
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- Assignments that are not typed will not be accepted.
 - Late Assignments will not be accepted without prior approval.
 - You may use the Internet as well as ChatGPT or other public tools as long as you place a written statement about their use in your assignment report

Problem A: Review of Chemical Engineering Fundamentals

Your first assignment after six months of training in the **UIC Specialty Chemicals Corporation** is to provide a material balance for a cyclohexane production unit and preliminary cost estimates of the utilities. A simple process flow diagram (shown in Figure A below) has been given to you by a mid-level engineer who has decided to leave the company. The data for this process are given below:

Benzene (BZ) Feed

Flow: 92.14 lbmole/hr

Temperature: 100 °F

Pressure: 15 psia

Hydrogen (H₂) Feed

Flow: 283.8 lbmole/hr

Temperature: 120 °F

Pressure: 335 psia

H₂ mole fraction : 0.975

N₂ mole fraction : 0.005

CH₄ mole fraction : 0.020

Charge Heater

Temperature: 300 °F

Pressure: 330 psia

Reactor

Temperature: 400 °F

Pressure: 315 psia

Benzene Conversion: 99.86%

H₂ to BZ molar ratio: 3.3

High Pressure (HP) Product Separator (Flash Drum)

Temperature: 120 °F

Pressure : 300 psia

Low Pressure (LP) Product Separator (Flash Drum):

Pressure : 15 psia

Your tasks

1. Complete the process flowsheet by adding the missing unit operations and provide a brief explanation for every unit you add.
2. Prepare the material balance in the format of the **ENG Calculations** of the **CHE Calculator**. Provide a brief explanation on how you calculate the effluent streams from each unit operation. Additionally provide a brief description of the numerical method(s) you have used to derive the material balance
3. Provide a preliminary cost of all utilities
4. Propose a method to reduce the utilities cost and do a simulation to calculate the cost reduction of your proposal

Data and Assumptions:

1. You may assume that the hydrocarbon mixtures with hydrogen presence in the process follow the Peng-Robinson equation of state
2. You may also assume that the K-values for the flash calculations can be calculated from Raoult's law or the Wilson equation
3. The fuel in the charge heater is natural gas
4. Utility cost data can be obtained from a variety of sources

Figure A: Hydrogenation of Benzene to Cyclohexane



